Testing of Statistical Hypothesis

- **Parameter:** Statistical constants of the population are known as parameter.
  For example: Population mean (μ), variance (σ^2) and correlation coefficient (ρ). Population parameters are generally unknown and estimated from the sample.
- **Sample:** A representative subset of the population is known as sample.
- **Sample Size:** Total number of the units in the sample is known as sample size.
- **Statistic:** Any function of sampling value is known as statistic.
  For example: if $x_1, x_2, x_3, \ldots, x_n$ be the sample observations from a population of size N, then $\bar{x} = \frac{x_1+x_2+x_3+\cdots+x_n}{n}$ is the statistic
- **Estimator:** The statistic used to estimate any population parameter is known as estimator.
- **Estimate:** An estimator is a random variate which takes different values from sample to sample. An estimator takes a particular value in a sample is known as its estimate.

Standard Error (SE): Standard deviation of sampling distribution of a statistic is known as standard error

$SE(\bar{x}) = \frac{\sigma}{\sqrt{n}}$

Importance of standard error:

i) Standard error play an important role in large sample theory and form the basis of testing of the statistical hypothesis

ii) Magnitude of SE gives an idea about precision of the estimate of population parameter. Precision of the estimate is reciprocal to the SE of the estimate.

iii) It gives probable limit within which the population parameter lies.

$$\bar{x} - \left(t_{\alpha/2}(n-1) \times SE\right) < \mu < \bar{x} + \left(t_{\alpha/2}(n-1) \times SE\right)$$

Example 7.3: Mean of 25 observation is 30 and SE is 2.1, find 95 % lower and upper limits.
Solution:

$\{30 - \left(t_{(0.05,24\ df)} \times 2.1\right) < \mu < 30 + \left(t_{(0.05,24df)} \times 2.1\right)\}$
$30 - 2.06 \times 2.1 < \mu < 30 + 2.06 \times 2.1$
$30 - 4.33 < \mu < 30 + 4.33$
$25.67 < \mu < 34.33$
Hence 95 % confidence limit for population mean is $25.67 < \mu < 34.33$

Statistical Hypothesis: Statistical hypothesis is some statement about the population or equivalently about the probability distribution characterizing a population which one want to verify on the basis of information available from the sample.

Statistical hypothesis is of two types

Test of statistical hypothesis: It is a two action decision problem after the sample observations has been obtained, two actions may be acceptance or rejection of the hypothesis.

Null hypothesis: The hypothesis which one want to test or the hypothesis of no difference. Null hypothesis is the hypothesis which is tested for possible rejection under the assumption that it is true. It is denoted by H_0.

$H_0: \mu = \mu_0$, or $H_0: \sigma^2 = \sigma_0^2$

Alternative hypothesis: The hypothesis which differ from null hypothesis is known as alternative hypothesis. It is denoted by H_1.

Alternative hypothesis may be any one of the following hypothesis

i) $H_1: \mu \neq \mu_0$, two tailed alternative hypothesis

ii) $H_1: \mu > \mu_0$, single tailed alternative hypothesis – right tailed

iii) $H_1: \mu < \mu_0$, single tailed alternative hypothesis – left tailed

One tailed test: A test of any statistical hypothesis where alternative hypothesis is single tailed (right tailed or left tailed) is known as one tailed test.

For example: $H_0: \mu = \mu_0$, $H_1: \mu > \mu_0$

or

$H_0: \mu = \mu_0$, $H_1: \mu < \mu_0$

Two tailed test: A test of any statistical hypothesis where alternative hypothesis is two tailed is known as two tailed test.

For example : $H_0: \mu = \mu_0$, $H_1: \mu \neq \mu_0$

Errors in testing of hypothesis: The decision to accept or reject the null hypothesis on the basis of information collected from sample may not always be true for the population parameter. There are four situations that may exist during any test procedure

True State	Decision from the sample	
	Reject H_o	**Accept H_0**
H_o True	**Wrong Decision (Type I Error)**	Correct Decision
H_o false	Correct Decision	**Wrong Decision (Type II Error)**

Type I Error: The error of rejecting H_0 when H_0 is true is known as Type I error. (or accepting H_1 when H_0 is true).

Type II error: The error of accepting H_0 when H_0 is false is known as type II error (or reject H_1 when H_0 is false).

Level of significance (α): Probability of type I error is known as level of significance or probability of rejecting H_0 when H_0 is true

Critical Region: The region of rejection of H_0 when H_0 is true is that region of outcome set where H_0 is rejected if the sample points fall in that region or a region in the sample space which amount to rejection of null hypothesis.

Critical value: A value which separate the acceptance region from the rejection region. It depends upon

i) Level of significance
ii) Alternative hypothesis whether it is two tailed or single tailed.

Degree of freedom: Number of independent parameters which make the test statistic.

e.g. If it is stated the sum of five number is 20. Suppose four numbers are 4, 3, 1 and 5, fifth number will be 20 - 13 =7. Here four numbers are independent and fifth one depends upon the value of these four number.

Procedure for testing of statistical hypothesis:

i) Set up null hypothesis (H_0)
ii) Set up alternative hypothesis (H_1)
iii) Select appropriate level of significance (α)
iv) Apply suitable test statistic
v) Conclusions.
   a) In two tailed alternative hypothesis, reject H_0 if |test statistic| > table value
   b) In right tailed alternative hypothesis, reject H_0 if test statistic > table value
   c) In left tailed alternative hypothesis, reject H_0 if test statistic < table value.

7.1.1 Application of Chi square test ($\chi 2$):

i) To test the goodness of fit

ii) To test the independence of attribute

7.1.2 Application of t test:

i) To test the hypothesis that population mean is equal to some given value

7.1.1: Chi square Test:

Chi square ($\chi 2$) variate: Suppose X is normally distributed with mean μ and variance σ^2 i.e. X~ $N(\mu, \sigma^2)$. Standard normal variate Z is defined as

$$Z = \frac{X - \mu}{\sigma}$$

i.e. $Z \sim N(\mu = 0, \sigma^2 = 1)$.

Square of a standard normal variate is known as Chi square ($\chi 2$) variate with 1 degree of freedom (df) i.e $Z^2 = \left(\frac{X-\mu}{\sigma}\right)^2$ is χ^2 variate with 1 df.

If $X_1, X_2, \ldots . Xi, \ldots, X_k$ are k independent random variable and $X_i \sim N(\mu_i, \sigma_i^2)$, sum of square of k independent standard normal variate is known as chi square distribution with k degree of freedom i.e.

$$\chi 2 = \sum \left(\frac{X-\mu}{\sigma}\right)^2 \sim \chi 2 \text{ with k degree of freedom.}$$

i) To test the goodness of fit :

It is a very powerful test for testing the differences between theoretical and experimental results. This test enables to find if the deviation of experimental results from theory is just by chance or is due to inadequacy of the theory. Suppose $O_1, O_2, O_3, \ldots . O_k$ be observed frequency in k classes and $E_1, E_2, E_3, \ldots . E_k$ be the expected frequency in corresponding classes according to the theory. To test the hypothesis that experimental results support the theory.

H_0: Experimental results support the theory

H_1: Experimental results does not support the theory

$\chi 2 = \sum \frac{(O-E)^2}{E}$, where O is observed frequency and E is expected frequency.

$\chi 2$ calculated value is compared with $\chi 2$ table value with k-1 df at 5 % level of significance. If $\chi 2$ calculated value is greater than $\chi 2$ table value, H_0 is rejected i.e. experimental results does not support the theory. If $\chi 2$ calculated value is less than $\chi 2$ table value, H_0 is not rejected i.e. experimental results support the theory.

Note: If any class frequency is less than 5, that frequency combined either with preceding or succeeding class and df is reduced by number of classes pooled.

Conditions for validity of $\chi 2$ test of goodness of fit:

a) Total frequency should be greater than 50

b) No class frequency should be less than 5

c) Sample observations should be independent

d) Sum of observed frequency should be equal to sum of expected frequency

Example 7.6: A survey was conducted with 1600 students and number of students from 4 regions Northern, Western, Eastern and southern were 875, 323, 297 and 105 respectively. Test the hypothesis that students from these regions are in the ratio of 9:3:3:1.

Class	O	E	(O – E)	$(O - E)^2$	$\frac{(O-E)^2}{E}$
AB	875	900	-25	625	0.69
Ab	323	300	23	529	1.76
aB	297	300	-3	9	0.03
ab	105	100	5	25	0.25
Total	1600	1600	0		2.73

H_0: Number of students in four regions are distributed in the ratio of 9:3:3:1
(or experimental results support the theory)

H_1: Number of students in four regions are not distributed in the ratio of 9:3:3:1
(or experimental results does not support the theory)

Total frequency = 875 + 323 + 297 + 105 =1600

Sum of ratio = 9 + 3 + 3 + 1 = 16

Expected frequency in any class = $\frac{\text{Proportion in the class}}{\text{sum of proportion}} \times N$

where N is total frequencies

Expected frequency in first class = $\frac{9}{16} \times 1600 = 900$

Expected frequency in second class = $\frac{3}{16} \times 1600 = 300$

Expected frequency in third class = $\frac{3}{16} \times 1600 = 300$

Expected frequency in forth class = $\frac{1}{16} \times 1600 = 100$

$\chi 2 = 2.74$

$\chi 2$ calculated value (2.74) is smaller than $\chi 2$ table value with 3 df (7.81), H_0 is not rejected i.e. experimental results support the theory and students from different regions are distributed in the ratio of 9:3:3:1 .

Example 7.8: Sale of an item during different days of week are 420, 415, 380, 375, 420 and 390 units. Test the hypothesis that sale during different days of week is equally distributed.

Days	**O**	**E**	**(O – E)**	**$(O-E)^2$**	**$\frac{(O-E)^2}{E}$**
Monday	420	400	20	400	1.00
Tuesday	415	400	15	225	0.56
Wednesday	380	400	-20	400	1.00
Thursday	375	400	-25	625	1.56
Friday	420	400	20	400	1.00
Saturday	390	400	-10	100	0.25
Total	2400	2400	0	2150	5.37

H_0: Sale of item is same in all days of a week

H_1: Sale of item is not same in all days of a week

Total frequency = 420 + 415 + 380 + 375 + 420 + 390 = 2400

Sum of ratio = 1+1+1+1+1+1 = 6

Expected frequency in each day = $\frac{1}{6} \times 2400 = 400$

$\chi 2 = 5.37$

or

$\chi 2 = \frac{2150}{400} = 5.37$

$\chi 2$ calculated value (5.37) is less than $\chi 2$ table value with 5 df (11.07), H_0 is not rejected i.e. experimental results support the theory and sale of item is equally distributed during different days of the week.

ii) To test the independence of attribute

Suppose there are two characters A and B. Character A is divided into r classes A_1, A_2, A_3,…A_i…A_r and character B is divided into s classes B_1, B_2, B_3,…B_j… B_s. Various cell frequencies will be expressed in following r × s contingency table.

Classes	**B_1**	**B_2**	**B_3**	**…**	**B_j**	**…**	**B_s**	**Total**
A_1	A_1B_1	A_1B_2	A_1B_3	…	A_1B_j	…	A_1B_s	**A_1.**

A_2	A_2B_1	A_2B_2	A_2B_3	…	A_2B_j	…	A_2B_s	**$A_{2.}$**
A_3	A_3B_1	A_3B_2	A_3B_3	…	A_3B_j	…	A_3B_s	**$A_{3.}$**
. . .	. . .	. . .	. . .	. . .	. . .	. . .	. . .	. . .
A_i	A_iB_1	A_iB_2	A_iB_3	…	A_iB_j	…	A_iB_s	**$A_{i.}$**
. . .	. . .	. . .	. . .	. . .	. . .	. . .	. . .	. . .
A_r	A_rB_1	A_rB_2	A_rB_3	…	A_rB_j	…	A_rB_s	**$A_{r.}$**
Total	**$B_{.1}$**	**$B_{.2}$**	**$B_{.3}$**	**…**	**$B_{.j}$**	**…**	**$B_{.s}$**	**N**

A_iB_j is number of individuals possessing both characters A and B

$A_{i.}$ is number of individuals possessing character A_i and

$B_{.j}$ is number of individuals possessing character B_j

N is total frequency.

$Ai. = A_iB_1 + A_iB_2 + A_iB_3 + \ldots A_iB_j + \ldots + A_iB_s$

$B_{.j} = A_1B_j + A_2B_j + A_3B_j + \ldots A_iB_j + \ldots + A_rB_j$

i=1,2,3,…, r and j = 1,2,3,…, s

$N = A_{1.} + A_{2.} + A_{3.} + \ldots + A_{i.} + \ldots + A_r$ or

$N = B_{.1} + B_{.2} + B_{.3} + \ldots + B_{.j} + \ldots + B_{.s}$

Here main objective is to test the hypothesis that two characters are independent to each other or not.

Probability that an individual possess the character A_i is

$$P(A_i) = \frac{A_{i.}}{N}$$

Probability that an individual possess the character B_j is

$$P(B_j) = \frac{B_{.j}}{N}$$

When character A and B are independent to each other, probability that a person possesses both characters A_i and B_j is

$$P(A_i \cap B_j) = P(A_i).\ P(B_j) = \frac{A_{i.}}{N} \times \frac{B_{.j}}{N}$$

Number of person possesses both characters A_i and B_j is

$$N.P(A_i \cap B_j) = N \times \frac{A_{i.}}{N} \times \frac{B_{.j}}{N} = \frac{A_{i.} \times B_{.j}}{N}$$

i.e. expected frequency of any cell is obtained by multiplying the row total of that cell with column total of that cell and divided by total frequency.

H_0: Both the characters are independent to each other

H_1: Both characters are not independent to each other

$\chi 2 = \sum \frac{(O-E)^2}{E}$, where O is observed frequency and E is expected frequency.

$\chi 2$ calculated value is compared with $\chi 2$ table value with (r-1) × (s-1) df at 5 % level of significance. If $\chi 2$ calculated value is greater than $\chi 2$ table value, H_0 is rejected i.e. both the characters are dependent to each other. If $\chi 2$ calculated value is less than $\chi 2$ table value, H_0 is not rejected i.e. both the characters are independent to each other.

Conditions for validity of $\chi 2$ test of independence of attribute:

a) Total frequency should be greater than 50
b) No cell frequency should be less than 5
c) Sample observations should be independent
d) Sum of observed frequency should be equal to sum of expected frequency

Example 7.13: In a survey of 500 students, data were classified into four computer brands and, types of their residence

Residence	**Types of Computer**			
	A	**B**	**C**	**D**
Rural	23	28	25	27
Urban	32	45	72	29
Metropolitan	56	48	79	36

Test the hypothesis that type of computer is independent of type of residence

Solution:

H_0: Type of computer is independent of residence type.

H_1: Type of computer is dependent to residence type.

Residence	**Types of Computer**				**Total**
	A	**B**	**C**	**D**	

Rural	23 **(23)**	28 **(25)**	25 **(36)**	27 **(19)**	103
Urban	32 **(40)**	45 **(43)**	72 **(63)**	29 **(33)**	178
Metropolitan	56 **(49)**	48 **(53)**	79 **(77)**	36 **(40)**	219
Total	111	121	176	92	500

E(23) = $\frac{103 \times 111}{500}$ = 23, E(28) = $\frac{103 \times 121}{500}$ = 25 , E(25) = $\frac{103 \times 176}{500}$ = 36 , E(27) = $\frac{103 \times 92}{500}$ = 19

E(32) = $\frac{178 \times 111}{500}$ = 40, E(45) = $\frac{178 \times 121}{500}$ = 43 , E(72) = $\frac{178 \times 176}{500}$ = 63 , E(29) = $\frac{178 \times 92}{500}$ = 33

E(56) = $\frac{219 \times 111}{500}$ = 49, E(48) = $\frac{219 \times 121}{500}$ = 53 , E(79) = $\frac{219 \times 176}{500}$ = 77 , E(36) = $\frac{219 \times 92}{500}$ = 40

where values on right hand side corner in parenthesis is expected frequency

O	**E**	**(O- E)**	**$(O - E)^2$**	**$\frac{(O-E)^2}{E}$**
23	23	0	0	0.00
32	40	-8	64	1.60
56	49	7	49	1.00
28	25	3	9	0.36
45	43	2	4	0.09
48	53	-5	25	0.47
25	36	-11	121	3.36
72	63	9	81	1.29
79	77	2	4	0.05
27	19	8	64	3.37
29	33	-5	25	0.76
36	40	-4	16	0.40
500	500	0		12.75

$\chi 2$ = 12.75

$\chi 2$ calculated value (12.75) is greater than $\chi 2$ table value with (3-1)×(4-1) = 6 df at 5 % level of significance (12.59), H_0 is rejected i.e. both the characters are dependent to each other i.e. brand of computer used depends upon type of redidence

2 × 2 Contingency table: Suppose two characters A and B, where A is divided into 2 classes A_1, A_2 and B is divided into two classes B_1,B_2,to test whether these two characters are independent or not ,following is the frequency

Classes	**A**		**Total**
B	**A_1**	**A_2**	
B_1	a	b	a+b
B_2	c	d	c+d

Total	a+c	b+d	N

where N = a + b + c + d

$$\chi 2 = \frac{N(a\times d - b\times c)^2}{(a+b)(c+d)(a+c)(b+d)}$$

Compare $\chi 2$ calculated value with $\chi 2$ table value with (2-1)×(2-1) = 1 df at 5 % level of significance. If $\chi 2$ calculated value is less than $\chi 2$ table value, H_0 is not rejected i.e. two characters are independent to each other, If $\chi 2$ calculated value is greater than $\chi 2$ table value, H_0 is rejected i.e. two characters are dependent to each other.

If any cell frequency is < 5 and if we pool the observation then df is (2-1)×(2-1) – 1 = 1-1 =0, i.e. no df is left for comparison. Hence Yate's correction is used when any cell frequency is < 5.

Yate's Correction: Add 0.5 to the cell whose frequency is less than 5 and adjust remaining frequencies so that row total and column total remains same. Suppose d < 5

Class	**A**		**Total**
	A_1	**A_2**	
B_1	a′= a + 0.5	b′= b - 0.5	a + b
B_2	c′= c - 0.5	d′ = d + 0.5	c + d
Total	a + c	b + d	N

$$\chi 2 = \frac{N(a'Xd' - b'Xc')^2}{(a+b)(c+d)(a+c)(b+d)}$$

Compare $\chi 2$ calculated value with $\chi 2$ table value with (2-1)×(2-1) = 1 df at 5 % level of significance.

If we do not add 0.5 to the cell, alternative formula to compute $\chi 2$ is

$$\chi 2 = \frac{N\left(|aXd - bXc| - \frac{N}{2}\right)^2}{(a+b)(c+d)(a+c)(b+d)}$$

Example 7.15: In a survey of 150 students, 69 were rural and Tall, 23 were rural and small, 47 were urban and tall, 11 were urban and small. Test the hypothesis that height is independent of area.

Solution:

Area	**Height**		**Total**
	Tall	**Small**	
Rural	69	23	92
Urban	47	11	58

Total	116	34	150

H_0: Height is independent of area
H_1: Height is dependent of area

$$\chi 2 = \frac{150(69\times11-23\times47)^2}{92\times58\times116\times34}$$

$$= \frac{150(759-1081)^2}{92\times58\times116\times34} = \frac{150(-322)^2}{92\times58\times116\times34} = \frac{150\times103684}{92\times58\times116\times34} = 0.74$$

$\chi 2$ calculated value (0.74) is less than $\chi 2$ table value (3.84) at 1 df at 5 % level of significance. H_0 is not rejected i.e. height is independent of area.

Example 7.16: Out of 350 rose plants,169 were having flat leaf and white colour flower, 123 were flat leaf and red colour flower, 47 were curled leaf and white flower, 11 were curled leaf and red flower. Test the hypothesis that colour of flower is independent of shape of leaf.

Solution:

Shape of leaf	**Colour of flower**		**Total**
	White	**Red**	
Flat	169	123	292
Curled	47	11	58
Total	216	134	350

H_0: Colour of flower is independent of shape of leaf

H_1: Colour of flower is dependent of shape of leaf

$$\chi 2 = \frac{350(169\times11-123\times47)^2}{292\times58\times216\times134}$$

$$= \frac{350(1859-5781)^2}{292\times58\times216\times134} = \frac{350(-3922)^2}{292\times58\times216\times134} = \frac{350\times15382084}{292\times58\times216\times134} = 10.98$$

$\chi 2$ calculated value (10.98) is greater than $\chi 2$ table value (3.84) at 1 df at 5 % level of significance . H_0 is rejected i.e colour of flower depends upon shape of leaf.

Example 7.17: Two types of plantation were raised under two irrigation systems. Test the hypothesis that types of plantation is independent of irrigation system.

Type of irrigation	**Plantation**		**Total**
	Monoculture	**Mixed**	
Unirrigated	67	23	90
Irrigated	27	3	30

Total	94	26	120

H_0: Plantation is independent of irrigation

H_1: Plantation is dependent of irrigation

Since one of cell frequency is less than 5, i.e. 3, add 0.5 to 3 and adjust remaining frequencies so that row and column total remain same.

Type of irrigation	**Plantation**		**Total**
	Monoculture	**Mixed**	
Unirrigated	67.5	22.5	90
Irrigated	26.5	3.5	30
Total	94	26	120

$$\chi 2 = \frac{120(67.5\times3.5-22.5\times26.5)^2}{90\times30\times94\times26}$$

$$= \frac{120(236.25-596.25)^2}{90\times30\times94\times26} = \frac{120(-360)^2}{90\times30\times94\times26} = \frac{120\times129600}{90\times30\times94\times26} = 2.36$$

Alternative method: If we do not add 0.5 to the cell

$$\chi 2 = \frac{120(|67\times3-23\times27|-\frac{120}{2})^2}{90\times30\times94\times26}$$

$$= \frac{120(|201-621|-60)^2}{90\times30\times94\times26} = \frac{120(|-420|-60)^2}{90\times30\times94\times26} = \frac{120(360)^2}{990\times30\times94\times26} = \frac{120\times129600}{90\times30\times94\times264} = 2.36$$

$\chi 2$ calculated value (2.36) is less than $\chi 2$ table value (3.84) at 1 df at 5 % level of significance. H_0 is not rejected i.e type of plantation is independent of irrigation type.

7.1.2 t test :

Student's t: If $x_1, x_2, x_3, \ldots, x_n$ be a random sample of size n from a normal population with mean μ and variance σ^2, then student t is

$t = \frac{\bar{x}-\mu}{\frac{s}{\sqrt{n}}}$ follows t distribution with n-1 df

where $\bar{x} = \frac{\sum_{i=1}^{n} x_i}{n}$ and

$s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2$

Fisher't: It is defined as ratio of standard normal variate to the square root of an independent chi square variate divided by its degree of freedom. If Z is a standard normal variate and $\chi 2$ is an independent chi-square variate with n df, then

Fisher $t = \frac{Z}{\sqrt{\frac{\chi^2}{n}}}$ and follows Fisher-t distribution with n df. where $Z = \frac{X-\mu}{\sigma}$

i) To test the hypothesis that population mean is equal to some given value:

Suppose a sample $x_1, x_2, \ldots, x_n$ of size n is drawn from the population whose mean is equal to μ_0.

H_0: $\mu = \mu_0$

H_1: $\mu \neq \mu_0$

$t = \frac{\bar{x} - \mu_0}{\frac{s}{\sqrt{n}}}$ follows t distribution with n-1 df

where $\bar{x} = \frac{\sum_{i=1}^{n} x_i}{n}$ and

$s^2 = \frac{1}{n-1} \sum_{i=1}^{n} (x_i - \bar{x})^2$

Assumptions:

i) Parent population from which the sample have been drawn is normally distributed

ii) Sample observations should be independent

iii) Population variance is unknown

Example 7.26: 10 students were selected randomly from a population whose weight is 87, 69, 58, 74, 63, 77, 90, 78, 73 and 81 kg. Test the hypothesis that population mean is 68 kg.

Student number	**We weight (X)**	$(\mathbf{X - 75})$	$(\mathbf{X - 75})^2$	$\mathbf{X^2}$
1	87	12	144	7569
2	69	-6	36	4761
3	58	-17	289	3364
4	74	-1	1	5476
5	63	-12	144	3969
6	77	2	4	5929
7	90	15	225	8100
8	78	3	9	6084
9	73	-2	4	5329
10	81	6	36	6561
Total	750	0	892	57142

H_0: $\mu = 68$

H_1: $\mu \neq 68$

Mean $\bar{x} = \frac{750}{10} = 75$

$s^2 = \frac{1}{10-1} \times 892 = 99.11$

or

$s^2 = \frac{1}{9}[57142 - \frac{750^2}{10}]$

$= \frac{1}{9}(57142 - 56250) = \frac{1}{9} \times 892 = 99.11$

$s = \sqrt{99.11} = 9.96$

$t = \frac{75 - 68}{\frac{9.96}{\sqrt{10}}} = \frac{7}{3.15} = 2.22$

t table value at 9 df is 2.26 . Since t calculated value (2.22) is less than 2.26, H_0 is not rejected, i.e. population mean is equal to 68.

7.1.4 Large Sample Test:

When sample size is greater than 30, large sample test is used.

i) Test of significance of single mean

Suppose a sample of size n is drawn from the population with mean μ_0 and variance σ^2. To test the hypothesis that sample have been drawn from the population having mean μ_0. i.e. no significant difference between sample mean $\bar{x}$ and population mean μ.

$H_0: \mu = \mu_0$

$H_1: \mu \neq \mu_0$

Test statistic $Z = \frac{\bar{x} - \mu_0}{\frac{\sigma}{\sqrt{n}}}$

At 5 % and 1 % level of significance table value is 1.96 and 2.58.

If | Z Calculated| > table value H_0 is rejected. i.e. difference between sample mean $\bar{x}$ and population mean μ is significant.

Example 7.36: A sample of 100 students gave mean weight as 55. Test the hypothesis that it is drawn from a population having mean weight as 60 with standard deviation 19.

Solution: n = 100 , $\bar{x} = 55, \mu_0 = 60$, $\sigma = 19$

$H_0: \mu = 60$

$H_1: \mu \neq 60$

$Z = \frac{55 - 60}{\frac{19}{\sqrt{100}}} = \frac{-5}{1.9} = -2.63$

Calculated value is greater than table value (1.96), H_0 is rejected, i.e sample has been drawn from the population having mean not equal to 60.

Example 7.37: A sample of 50 students gave mean weight as 55. Test the hypothesis that it is drawn from a population having mean weight as 60 with standard deviation 19.

Solution: n = 50 , $\bar{x} = 55, \mu_0 = 60$, $\sigma = 19$

$H_0 : \mu = 60$

$H_1 : \mu \neq 60$

$Z = \frac{55-60}{\frac{19}{\sqrt{50}}} = \frac{-5}{2.69} = -1.86,$

Calculated value is smaller than table value (1.96), H_0 is not rejected, i.e sample has been drawn from the population having mean = 60.

ii) **Test for single proportion**

Suppose there are n independent trial and each trial is having constant probability of success say P. if X is number of successes in n trial, then

Proportion of success (p) = $\frac{X}{n}$

SE(p) = $\sqrt{\frac{PQ}{n}}$

We want to test the hypothesis that sample have been drawn from the population having proportion P_0

H_0: P = P_0

H_1: P ≠ P_0

Test statistic Z = $\frac{p - P}{\sqrt{\frac{PQ}{n}}}$

If | Z Calculated| > table value H_0 is rejected.

Example 7.41: In a random sample of 1000 students , 75 were girl. Is this sample drawn from the population having proportion of girl as 0.1.

Solution:

H_0: P = 0.1

H_1: P ≠ 0.1

Proportion of girl (p) in the sample

p = $\frac{75}{1000}$ = 0.075

n = 1000, P = 0.1, Q = 1−0.1 =0.9,

Test statistic Z = $\frac{0.075-0.10}{\sqrt{\frac{0.1\times0.9}{1000}}} = \frac{-0.025}{\sqrt{0.00009}} = \frac{-0.025}{0.009487}$ = -2.64

| Z Calculated| value > is greater than table value (1.96), H_0 is rejected, i.e the proportion of girl in the population is not equal to 0.1

Example 7.42: In a random sample of 500 students of BCA, 225 students are boys and rest are girls. Test the hypothesis that boys and girls are in equal proportion in the university.

Solution:

H_0: P = 0.5

H_1: P ≠ 0.5

Proportion of boys (p)

p = $\frac{225}{500}$ = 0.455

n = 500, P = 0.5, Q = 1− 0.5 =0.5,

Test statistic Z = $\frac{0.45-0.50}{\sqrt{\frac{0.5\times0.5}{500}}} = \frac{-0.05}{\sqrt{0.0005}} = \frac{-0.05}{0.022361} = -2.24$

Calculated value is greater than table value (1.96), H_0 is rejected, i.e the proportion of boys and girls is not same.